

Windows Fundamental (SOC Focus)

Objective

To understand how Windows logs work, where events are stored, and which logs SOC analysts rely on for threat detection.

1. What I Learned Today

Windows Event Viewer: The Windows Event Viewer is a built-in tool that records all system, security, and application events. SOC analysts use it to detect:

- Failed login attempt
- Unauthorized login
- Malware execution
- System warnings and errors
- Suspicious process creation

2. Key Windows Log Categories

- Security logs : It contains authentication activity, access attempts and privilege use. It is used in detecting - Brute force attacks, Unauthorized logins and malicious process creation.
- Application logs : It stores event logs generated by programs running on windows
- System logs: It stores OS-level events such as startup,shutdown and service failures.
- Sysmon logs(System Monitor): It is part of the sysinternal suite. It is a powerful tools that keep tracks of :
 - i. Process creation
 - ii. File creation
 - iii. Network connections
 - Iv. Registry changes

Sysmon provides high-precision logs that SIEM tools rely on.

3. Critical Events IDs SOC Analysts Must Know

- 4624 - successful login

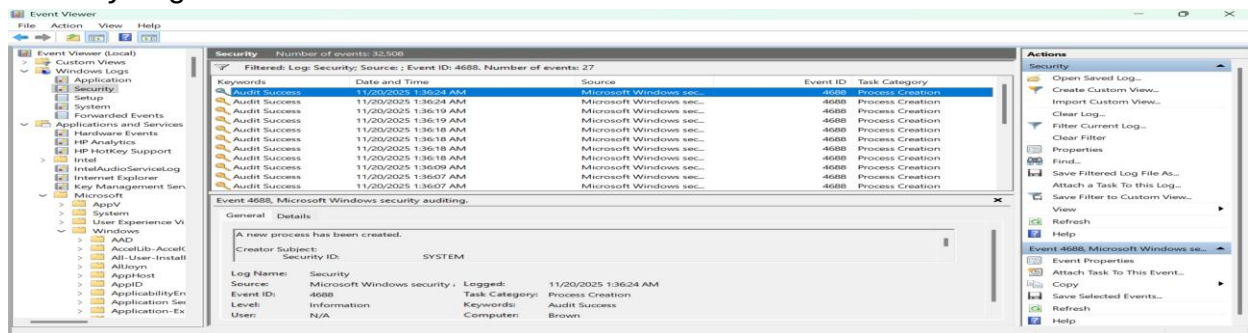
- 4625 - Failed login attempts: It can be used to detect - Brute force attempts, Wrong password attempts, Unauthorized account usage.
- 4688 - Process creation: It shows which program ran, who ran it, parent process and command-line arguments.
- 4672 - Special privileges assigned. It detects privilege escalation.
- 4720 - New account created. Detects attacker account creation.

4. Summary

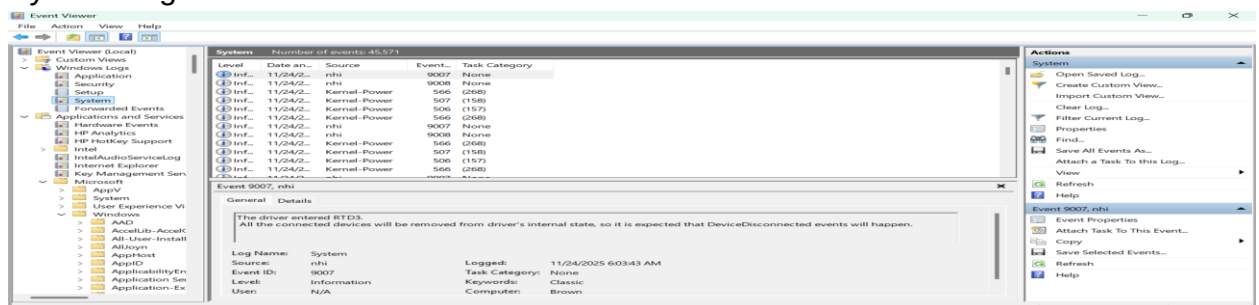
Windows logs are essential for detecting suspicious activity inside an organization. SOC analysts rely heavily on Security Logs, System Logs, Application Logs, and Sysmon logs to investigate potential threats. Critical event IDs such as 4624 (successful login), 4625 (failed login), and 4688 (process creation) provide visibility into user authentication and program execution. Understanding these logs allows analysts to detect brute force attacks, privilege escalation, malware execution, and unauthorized system access.

5. Screenshots

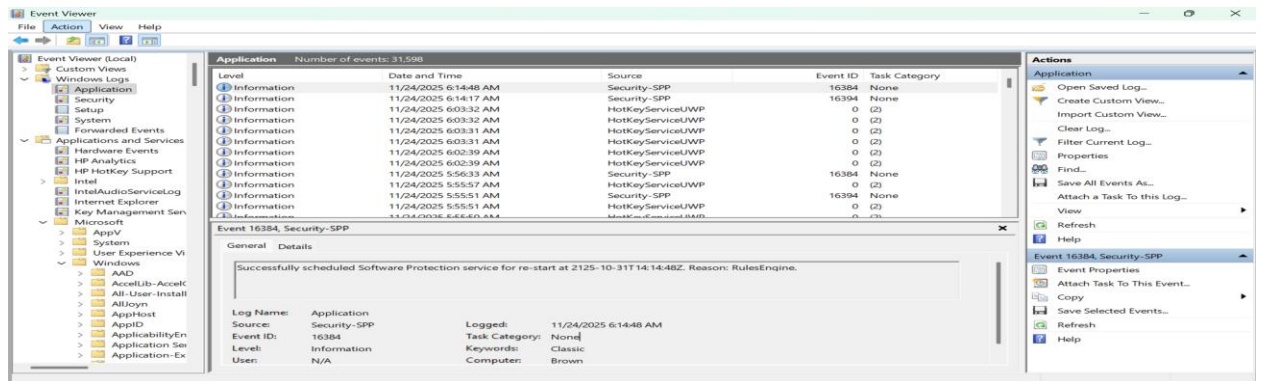
A. Security Logs



B. System Logs



C. Application Logs



D. Process Creation Logs(Sysmon)

